

Interview Protocol

Field	Value
Study type	Semi-structured expert interviews
Purpose	Validate conceptual clarity (RQ1) and practical applicability (RQ2) of the taxonomy
Number of participants	5
Interview format	Individual, one-to-one
Duration	Approximately 60 minutes per interview
Materials provided	Taxonomy slides (available in repository "ClassificationEvaluationMeetings.pdf")
Analysis method	Thematic analysis

Recruitment Criteria

Participants were recruited purposively to ensure breadth of relevant expertise. The following criteria were applied:

- Background in at least one of: Human-Robot Teamwork, Human Factors, or Requirements Engineering.
- Active or recent professional engagement with human-robot teaming systems in research, industry, or regulatory contexts.
- Sufficient domain knowledge to evaluate both the conceptual structure of the taxonomy and its practical applicability to real systems.

Five participants were recruited spanning three sectors:

- 2 academics with research backgrounds in human-robot teamwork and/or requirements engineering.
- 2 engineers from large robotics engineering companies.
- 1 specialist from a space agency.

Questions

Each interview was conducted using a structured slide deck to ensure consistency across participants. The slides are publicly available in the project repository. The interview covered two broad areas aligned with the research questions:

RQ1 Conceptual Validity

Participants were guided through each of the six high-level taxonomy categories in turn and asked:

- Is this category clearly and distinctly defined?
- Does the rationale for the subcategories make sense? Are the subcategories well-placed within this category?
- Does this category capture the intended class of human-robot teamwork requirements?
- Are there requirements you would expect to find here that are missing?

RQ2 Practical Applicability

Participants were asked to relate the taxonomy to real human-robot teaming systems they were familiar with:

- Can you identify gaps or missing coverage when you think about systems you have worked on?
- How useful would this taxonomy be for requirements elicitation on a new human-robot system?
- How useful would it be for reviewing or auditing an existing specification for coverage?
- Are there any categories where you would expect to find many requirements in practice but find few here?

Interview Length

Interviews were conducted via video call (Teams) and were audio recorded for note taking purposes. Consent to record was obtained before commencement.

Coding Process

Interview discussions were analysed using thematic analysis. The analysis proceeded in the following stages:

1. Familiarisation: interview notes or transcripts were read in full to develop an initial understanding of the data.
2. Initial coding: segments relevant to taxonomy validity, category clarity, coverage gaps, and practical utility were identified and labelled.
3. Theme development: codes were grouped into the five themes reported in the paper: (1) Awareness vs Action; (2) Reduced Human-to-Robot Information and Negotiated Control; (3) Ergonomics and Human Variability; (4) Human Performance Monitoring Challenges; (5) Additional Categories.
4. Review and refinement: themes were reviewed against the full data set and refined to ensure they accurately represented participant responses.